

Executive Business Report

Sales Forecasting and Demand Intelligence Dashboard

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Executive Summary

This report presents a Sales Forecasting and Demand Intelligence Dashboard developed to improve inventory planning and sales forecasting. Historical sales data was analyzed to identify trends, forecast future demand, detect unusual sales patterns, and classify products into demand segments. The dashboard enables business users to make data-driven decisions for inventory management, demand forecasting, and stock optimization through an easy-to-use interactive interface.

1. Key Findings

- Sales generally increased over the years with noticeable seasonal variations.
- Technology products generated the highest revenue.
- Furniture sales showed higher fluctuations than other categories.
- Office Supplies maintained stable demand.
- Sales performance differed across East and West regions.
- Product demand varied significantly across sub-categories.

2. Three-Month Sales Forecast

The XGBoost model was selected as the best forecasting model and trained separately for each required business segment.

Segment	Month 1	Month 2	Month 3
Furniture	23,165	34,390	35,658
Technology	24,043	26,460	22,461
Office Supplies	26,713	26,576	26,610
West Region	17,766	19,020	29,882
East Region	21,492	22,405	22,671

Plain Language: These forecasts represent the most likely sales based on historical trends. Actual sales may vary because of promotions, seasonal demand, market conditions, and customer behavior.

3. Top Anomalies Detected

Isolation Forest and Z-Score techniques identified unusual sales periods caused by:

- Promotional or festive sales spikes
- Stock shortages leading to temporary sales drops
- Sudden demand fluctuations due to changing customer behavior

These anomalies help managers investigate exceptional business events before making inventory decisions.

4. Product Demand Segmentation

Products were grouped using K-Means clustering into four demand segments.

Demand Segment	Recommended Stocking Strategy
High Volume – Stable Demand	Maintain higher inventory levels
Growing Demand	Increase stock gradually based on demand
High Value – High Volatility	Maintain safety stock and monitor frequently
Low Volume – Stable Demand	Keep minimal inventory and replenish when required

5. Business Recommendations

1. Increase inventory for Technology products because they consistently generated high sales.
2. Maintain safety stock for Furniture products to reduce stock-out risk during demand fluctuations.
3. Use monthly XGBoost forecasts for inventory planning to reduce excess stock while improving product availability.

6. Risk / Limitation

The system relies on historical sales data and may not accurately predict unexpected events such as economic changes, competitor promotions, supply chain disruptions, or sudden market shifts. Forecasts should therefore support—not replace—business judgment.

Conclusion

The dashboard combines sales analysis, forecasting, anomaly detection, and product segmentation into a single decision-support system. It enables business users to improve inventory planning, optimize stock levels, and make informed decisions using reliable sales insights.

Live App Link: <https://salesforecasting-gosai-jeevika.streamlit.app/>